

MILGRAIN

The wearable intelligence pendant.

Product Overview & Technical Reference

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"Where fashion meets intelligence."

1. What Is Milgrain

Milgrain is a wearable AI pendant. You press it, speak naturally, and things happen — without taking your phone out. It is positioned as a life operating system, not a chatbot or a voice recorder.

The pendant is worn around the neck. It captures voice, transmits audio to a companion iOS app over BLE, which forwards it to a cloud backend for AI reasoning and action execution. The response comes back as speech. The phone stays in your pocket throughout.

The hardware is a 20–24 mm circular titanium pendant with a swappable shell system, connected via a conductive chain to a countermodule at the nape of the neck. The countermodule carries the battery and bone conduction output — it is the weight, the audio, and the health sensor platform. The pendant is the face — small, light, jewellery-first.

What Milgrain is

- A wearable personal assistant with real action capability.
- A hands-free, screen-light interface for daily life.
- A connector and action system for real services: Gmail, Calendar, Telegram, Uber, trains, food delivery, Apple Music, Maps, HomeKit and more.
- A memory system that learns your context over time.
- A proactive briefing surface — relevant context before you ask.
- A fashion-tech object with a swappable shell identity.
- A companion iOS app plus backend today — the pendant is the full product expression.

What Milgrain is not

- Not a phone replacement.
- Not a note-taker only.
- Not a generic AI chatbot wrapper.
- Not a camera-first product — there is no camera.
- Not passive surveillance. Press-to-talk is the core trust model: you initiate every interaction.

2. Identity & Brand

Milgrain is the brand and product name. The identity is built around a gold dot-M monogram — approximately 1,000 gold dots hex-packed on a cream ground, the same density and rhythm as milgrain jewellery edging, which is the namesake. The wordmark is set in Didot Regular, all-caps.

The brand sits in the intersection of luxury jewellery and precision engineering. The aesthetic is warm precision: deep space black environments, gold accents, cream typography, high-contrast editorial light.

The audience is inclusive and feminine-leaning luxury without feminine cliché. The design de-gadgets the product — it should read as jewellery first, technology second.

Logo	Dot-M monogram: ~1,000 gold dots, hex-packed on cream ground
Wordmark	Didot Regular, all-caps — MILGRAIN
Primary colour	Gold — #C9A84C
Background	Deep space black / cream
Typography	Georgia (display), Arial (body)
Positioning	Luxury + intelligence. Fashion-tech, not gadget-tech.
Tagline	"Where fashion meets intelligence."
Alt tagline	"One intelligence. Infinite expression."

3. Form Factor

3.1 Two-module architecture

Milgrain uses a split design: a small pendant at the front and a countermodule at the nape of the neck. They are connected by a conductive chain — the chain carries both power and data between the two modules. From the outside it looks like a normal necklace chain.

Pendant (front)	Mic, MCU, BLE — the visible jewellery piece. 20–24 mm diameter.
Countermodule (nape)	Battery, bone conduction transducer, health sensors — rests against the neck.
Connection	Conductive chain carrying power and signal between both modules.
Why split	Pendant stays small and light. Countermodule is a natural counterweight that keeps the pendant sitting flat. Bone conduction and health sensing from the nape is anatomically superior to any front-mounted path.

3.2 Physical dimensions

Pendant diameter	20–24 mm
Pendant material	Grade 5 titanium body, frosted Gorilla Glass 6 face
Countermodule	Sits at nape of neck — dimensions TBD, sized to house battery + sensors
Weight distribution	Countermodule acts as counterweight; pendant sits flat without pulling forward
Worn	Around the neck. Pendant at front, countermodule at back.
Chain	Conductive — carries power and signal. Included with purchase.

3.3 Core + shell architecture

The pendant body uses a two-part system. The core contains all electronics and never changes. The shell is the outer face and back — the visible piece — and can be swapped. This allows multiple finish options, designer collaborations without re-engineering electronics, and shell drops as cultural/revenue events.

3.4 Shell materials & finishes

Launch — Brushed Titanium	Grade 5 Ti, natural brushed. The hero finish.
Launch — PVD Noir	Grade 5 Ti, physical vapour deposition black coating.
Launch — PVD Gold	Grade 5 Ti, PVD gold coating.
Drop — Rose Gold	PVD rose gold, seasonal.

Drop — Designer collabs	UK jewellery designer partnerships, limited batch.
Face	Frosted Gorilla Glass 6 — scratch-resistant, diffused.
Shell price	£79–£89 per shell (sold separately after initial).

3.5 Interaction model

One primary interaction: press to talk. Press the pendant, speak, release. The pendant streams audio over BLE to the phone. The AI processes the request and responds through bone conduction at the nape. Haptics confirm the press. LED gives status feedback. Nothing is captured without the button press — no ambient listening, no camera.

3.6 Charging

The charger is the jewellery box. The pendant sits in a custom recess; Qi wireless charging powers it without cables. The box is portable enough for travel. This makes the object feel like jewellery, not a gadget, from the moment of unboxing.

4. Hardware & PCB

4.1 PCB specification

Two PCBs: one in the pendant (~20–24 mm circular) and one in the countermodule (larger, carries battery, bone conduction, and health sensors).

Pendant PCB layers	4-layer FR4
Pendant PCB diameter	~20–24 mm circular
Countermodule PCB	Dimensions TBD — sized to fit at nape with battery and sensors
Thickness	1.6 mm (both boards)
Soldermask	Black
Surface finish	ENIG (Electroless Nickel Immersion Gold)
Target fab	JLCPCB with PCBA
Chain connection	Conductive pads on both PCBs — chain carries power + data

4.2 Bill of materials

Component	Part / Target	Notes
Production MCU	Raytac MDBT50Q-512K (nRF52840)	Pendant PCB — BLE + processing, low power
Dev/prototype MCU	ESP32-S3 (Seeed XIAO nRF52840 Sense)	Prototype — easier bring-up, Arduino BSP
Microphone	Knowles SPH0645LM4H	Pendant PCB — I2S MEMS, high SNR
Touch IC	Infineon CY8CMBR3102	Pendant PCB — capacitive press through glass face
LED	Worldsemi WS2812B SMD 5050	Pendant PCB — RGB status / notifications
Haptic motor	AAC Technologies NFP-ELV1040	Pendant PCB — press confirmation haptics
IMU	Bosch BMI270	Pendant PCB — tap / wear detection, sleep motion
Bone conduction transducer	AAC Technologies AT23B46	Countermodule — nape/skull contact
Bone conduction amp	TI TPA6205	Countermodule — drives transducer
Heart rate / SpO2 / HRV	Maxim MAX30102	Countermodule — PPG optical at nape skin
Skin temperature	TI TMP117 (candidate)	Countermodule — high-accuracy $\pm 0.1^{\circ}\text{C}$

Charging IC	TI BQ25185	Countermodule — single-cell Li-ion, USB-C / Qi
Qi module	TDK WR202	Countermodule — wireless charging receiver
Battery	ATL 503040 — 500 mAh	Countermodule — bulk weight at nape

4.3 MCU rationale

Prototype uses an ESP32-S3 (Seeed XIAO nRF52840 Sense) because it is easy to bring up with the Adafruit BSP, Bluefruit library, and onboard PDM mic. Production switches to the nRF52840 for power: the ESP32 is too hungry for a 500 mAh battery with multi-day standby requirements.

4.4 Firmware — current prototype

The prototype firmware runs on the Seeed XIAO nRF52840 Sense. It streams 16 kHz / 16-bit PCM audio from the onboard PDM mic over BLE Nordic UART to the iOS companion app. Key implementation details:

- 8,192-sample ring buffer decouples PDM capture interrupt from BLE transmission — prevents corrupted audio from ISR/loop collisions.
- MTU 247 + data-length extension negotiated on connect — enables ~244-byte BLE payloads, sufficient to sustain the 256 kbps stream.
- Receives command strings from iOS app ("THINK", "DONE", "PING") and drives a non-blocking LED blink state machine in response.
- Restarts advertising immediately on disconnect, 30-second fast window then slow background advertising.

4.5 Bone conduction

The bone conduction transducer sits in the countermodule at the nape of the neck. The nape is anatomically well-positioned: the mastoid bone and occipital ridge are established bone conduction paths used by hearing aids and purpose-built bone conduction headphones. The countermodule rests against the neck by gravity and chain tension — no strap or clip needed.

The nape has consistent hairless skin contact, thermally stable for health sensing, and proximity to the carotid and jugular for strong PPG signal. It is a better health sensing site than the finger (Oura) or wrist (Apple Watch) for resting metrics.

Open item: acoustic transmission through clothing vs bare skin at the nape still needs to be physically measured in prototype.

4.6 Manufacturing path

PCB fabrication	JLCPCB — 4-layer, black soldermask, ENIG, circular pendant PCB
PCB assembly	JLCPCB PCBA with LCSC BOM where possible
CNC titanium shell	Xometry or Shenzhen precision supplier — one sample before batch
Gorilla Glass face	Custom circular cut, frosted

Conductive chain	Supplier TBD — contact resistance, current capacity, and flex under daily wear to be validated
Assembly	Manual for prototypes; DFM review before production run
Waterproofing	IP rating strategy not yet defined — pendant and nape module both need it

4.7 Hardware open items

- Gerbers not yet produced — two PCBs needed. KiCad design is next step.
- Conductive chain spec not finalised — contact resistance, current capacity, flexibility.
- PCBA and CNC quotes not obtained — unit economics are estimates until real quotes.
- Countermodule physical dimensions and enclosure not designed — must fit battery + sensors at nape comfortably.
- Bone conduction coupling at nape (clothing vs bare skin) not physically tested.
- Thermal and power consumption measurements not taken.
- Waterproofing / IP rating strategy not defined.
- DFM review not completed.
- CoreBluetooth BLE pendant bridge not yet present in iOS app — firmware is written, iOS side is pending.

5. Software Architecture

Milgrain is a multi-surface stack: a Node/Express backend on Google Cloud Run, a PWA, a native SwiftUI iOS app, a Supabase Postgres database, the Gemini model family for AI, and a connector/action system for real services. A standalone MCP server provides an external-agent bridge.

5.1 Signal path — voice request

User presses pendant !' BLE audio streams to iOS app !' app forwards to backend /process-audio via HTTPS !' Gemini transcribes audio !' conversation history, memories, preferences, and connector context load from Supabase !' Gemini reasons and generates response !' if an <action> block appears, the connector system executes it !' results fed back to Gemini for natural summarisation !' Gemini TTS generates speech !' SSE stream returns text + audio chunks to app !' audio plays through bone conduction at nape. Phone stays in pocket.

5.2 Backend

Runtime	Node.js, Express 5.1
Realtime	WebSocket via ws — /realtime-voice, /companion-live
Database	Supabase JS client !' Postgres
AI SDK	@google/generative-ai + @google/genai
Deploy target	Google Cloud Run (Docker container)
Proactive jobs	Cloud Run Job, Cloud Scheduler, every 15 min
Current URL	https://oxy-151340634966.europe-west2.run.app

5.3 iOS app (SwiftUI)

Screens	Chat · Today (briefings) · History · Connectors · Memory · Settings
Auth	Keychain-backed session storage, Bearer token
Streaming	SSE parsing, base64 WAV audio playback
Local actions	Messages, FaceTime, Apple Music, HealthKit, Reminders, Calendar, Contacts, Maps, Places
Context sync	POST /native/context — location, health snapshot, capabilities, settings
Permissions	Microphone, Location (always), Contacts, Health, Music, Calendar, Reminders, Photos
Pendant BLE	CoreBluetooth — in progress

5.4 Database schema

users	id, user_id, password_hash, timestamps
conversations	role, content, timestamp — per-user message history
memories	Extracted/manual personal facts, source type
action_log	Action JSON, status, error, timestamp
connectors	Connector ID, enabled, OAuth tokens (JSONB), per-user unique
preferences	Key/value — user preferences + proactive sent-state keys
devices	Platform, push token, timezone — APNs device registry
native_context	Latest location, health snapshot, capabilities, settings (JSONB)
briefings	Proactive briefing feed — kind, title, body, source, metadata, read

5.5 AI model layer

Primary chat model	gemini-3-flash-preview
Image model	gemini-2.5-flash-image
STT	Gemini — audio transcription via /process-audio
TTS	Gemini streaming TTS, sentence-level chunks, PCM !' WAV. Default voice: Aoede.
Search grounding	Enabled for factual/current queries. Disabled for personal/action queries.
Prompt caching	Warm cache + 5-min user context cache to reduce Supabase reads and latency
Gemini Live	WebSocket realtime voice path — /realtime-voice and /companion-live implemented, benchmarking pending

6. Connectors & Actions

The connector system is the core value proposition. Milgrain does not just talk — it acts. Each connector handles auth, token refresh, API calls, and deep-link fallbacks independently. Actions carry explicit risk levels and confirmation policies.

6.1 Connector registry

Connector	Actions	Auth	Status
Google	send_email, get_emails, search_emails, create_calendar_event, get_calendar_events	OAuth 2.0, per-user tokens	Live
Telegram	send_telegram, get_telegram_contacts	Telegram User API, phone + 2FA	Live
Uber	book_uber	Deep links + geocoding	Live
Uber Eats	order_uber_eats	Deep links	Live
Deliveroo	order_deliveroo	Deep links	Live
Netflix	search_netflix_title, add_to_netflix_list	App/web links	Live
Trainline	search_trains, station_board	TransportAPI + Trainline URLs	Live
Maps	find_place, get_directions, plan_trip	Google Places + Apple Maps fallback	Live
Bolt	book_bolt	Deep links	Live
HomeKit	smart_home	Home framework on-device	Live via native
Apple Wallet	wallet actions	PassKit on-device	Live via native
WhatsApp	send_whatsapp	Deep link + contact resolution	Live
Spotify	play_music	Spotify Web API	MCP server path
Home Assistant	smart_home	HA REST API	MCP server path
Twilio	make_call, SMS	Twilio API	MCP server path

6.2 Native iOS actions (on-device, no server needed)

- iMessage / SMS — MessageUI framework
- FaceTime / phone calls — URL scheme
- Apple Music playback, add to library/playlist — MusicKit / MediaPlayer
- HealthKit queries — steps, heart rate, sleep
- Reminders — create, read via EventKit
- Calendar events — create, read via EventKit

- Contacts — search, hint resolution
- Local place search — MapKit nearby search

Native actions execute on-device and report back to /native/local-action so Milgrain memory and history stay coherent.

6.3 Action safety model

Review-required (high risk)	send_email, send_telegram, send_whatsapp, make_call, order_uber_eats, order_deliveroo, book_uber
Execute directly (low risk)	get_emails, search_emails, get_calendar_events, find_place, get_directions, search_trains, play_music, search_netflix_title
Creative	generate_visual, create_diagram, create_presentation
Memory	forget_memory — user-initiated, no review required

7. Proactive Intelligence

Milgrain does not wait to be asked. A Cloud Run Job (every 15 minutes) runs a proactive sweep across registered users, checking native context — location, health, calendar, email — and generates contextual briefings that appear in the Today tab before the user opens the app.

7.1 Briefing types

Wake briefing	6–10 AM. Calendar summary, email digest, travel plan for the day.
Midday briefing	12–2 PM. Afternoon context, any follow-ups, relevant news.
Evening briefing	5–8 PM. End-of-day summary, reminders for tomorrow.
Health alert	Any time. Unusually low heart rate flagged from HealthKit / nape sensor.
Location food reminder	Near home in evening + active autonomy level !' food suggestion.
Action follow-up	Failed actions needing reconnection or permission resurface here.

7.2 Autonomy levels

Users set an autonomy level that gates how proactive Milgrain is. At Low/Quiet, interval briefings are suppressed. At High/Bold, location-aware reminders and food suggestions are active. Health alerts and location reminders have separate permission flags.

7.3 Context sources

- Google Calendar — events, today's schedule
- Gmail — latest email snippets
- HealthKit / nape sensor — heart rate, steps, sleep
- CoreLocation — current location from iOS app
- Reminders — pending items via EventKit
- Gemini Search grounding — weather, news, travel conditions

8. Use Cases & Demo Scenarios

These are the scenarios that convert sceptics to users. Each keeps the phone in the pocket and shows real execution.

8.1 Communication

- "Text Mum I'm on my way." !' iMessage sent via native action.
- "WhatsApp James the flight is delayed." !' WhatsApp deep link with pre-filled message.
- "Send a Telegram to the group chat: running five minutes late." !' Telegram User API, review, sent.
- "Read me my last three emails." !' Gmail, summarised aloud.
- "Reply to Lena saying I'll confirm Thursday." !' Gmail send, review required.

8.2 Transport & navigation

- "What train can I take to London before 9?" !' TransportAPI, spoken timetable.
- "Book me an Uber home." !' Uber deep link, current location + home address.
- "Find the nearest gym and open directions." !' MapKit / Google Places + Apple Maps.
- "How long to get to Waterloo right now?" !' Maps directions with live traffic.

8.3 Daily life

- "Remind me to call Sarah tomorrow at 10." !' Reminders via EventKit.
- "What did I say about my exams?" !' Memory lookup, spoken.
- "Add this song to my workout playlist." !' MusicKit.
- "Order me a pizza from Deliveroo." !' Deliveroo deep link, review shown.
- "What's on my calendar today?" !' Google Calendar, spoken summary.

8.4 Health

- "How was my sleep last night?" !' HealthKit / nape sensor HRV + sleep staging.
- "What's my resting heart rate this week?" !' Nape sensor PPG data, trend spoken.
- "Am I ready to train today?" !' Readiness score from HRV + sleep + temperature.

8.5 Creative & information

- "Generate an image of a minimalist Tokyo apartment at night." !' Gemini image generation.
- "Who won the Wimbledon final?" !' Gemini with Search grounding.
- "Make me a presentation outline for the pitch on Thursday." !' Structured deck artifact.

9. Unit Economics & Pricing

9.1 Hardware pricing

Pendant — standard	£349
Kickstarter early bird	£299
Collector tier	~£449 (extra shell + subscription period)
Shell (sold separately)	£79–£89 per shell

9.2 Subscription pricing

Milgrain Plus — monthly	£14.99/month (pricing ceiling)
Milgrain Plus — annual	£129–£149/year (< £12.50/month equivalent)
First month	Free with pendant purchase
Kickstarter inclusion	3–12 months included depending on tier
Free tier	Basic setup, limited assistant use, onboarding, connector demos

9.3 Hardware unit cost estimates

These are estimates. Real PCBA and CNC quotes have not been obtained. Do not use for investor modelling without actual quotes.

Volume	Est. unit cost	Margin at £349
100 units	~£240–£276	£73–£109
1,000 units	~£188	~£161
5,000 units	~£145	~£204

9.4 Subscription contribution at £14.99

Channel	Gross/user	After Apple cut	Est. contribution
Web (annual push)	£14.99/mo	Near full	~£9/mo after API
App Store	£14.99/mo	~£10.49 (30% cut)	~£5–6/mo after API
Blended 60% web / 40% App Store	—	~£13.19	~£7–9/mo after API

9.5 Revenue milestones

Subscribers	Est. monthly contribution
500	£3,500–£4,500
1,000	£7,000–£9,000
5,000	£35,000–£45,000
10,000	£70,000–£90,000

Hardware margin and shell drops matter more at this subscription price. A £204 margin per pendant at 5,000 units is £1M+ gross hardware contribution. Shells at £79–£89 with near-zero BOM cost are high-margin recurring revenue.

10. Competitive Landscape

The AI wearable category is moving quickly. Milgrain's differentiated combination is the moat — no single competitor holds all of these simultaneously.

Product	Form	Camera	Actions	Health	Fashion
Milgrain	Pendant (20–24mm)	None	Full connector system	HR, HRV, SpO2, temp, sleep	Yes — shell system
Humane AI Pin	Chest clip	Yes	Limited	None	No
Omi	Pendant/clip	No	Companion-focused	None	No
Meta Ray-Ban	Glasses	Yes	Limited	None	Partial
Oura Ring	Ring	No	None	Full health focus	Some
Taya	Pendant	No	Early	Some	Partial
Clicky	Pendant	No	Limited	None	No

Milgrain's differentiated combination

- Pendant form factor — wearable, not a gadget clip.
- No camera — trust by design.
- Press-to-talk — agency model users understand and trust.
- Bone conduction from nape — hands-free audio without earbuds.
- Full action execution — real things happen, not just answers.
- Health sensing from nape — HR, HRV, SpO2, temperature, sleep.
- Memory system — context that builds over time.
- Proactive briefings — relevant context before you ask.
- Fashion shell system — repeat revenue, designer collabs, cultural drops.
- Native iOS app — deep platform integration, on-device actions.

11. The Six-Layer Moat

1. Hardware

A 20–24 mm titanium pendant with a conductive chain to a nape countermodule carrying battery, bone conduction, and health sensors. This specific object — small, jewellery-grade, split-architecture — does not exist elsewhere. Hardware takes 12–18+ months to credibly copy.

2. Media & Brand

Fashion object first impression. Editorial renders. Shell drops as cultural moments. Designer collaborations. Jewellery-box unboxing. Distribution advantages that compound over time and cannot be cloned by shipping a product.

3. Deep Tech

Gemini Live realtime voice, prompt caching, context caching, deterministic intent routing, action contracts with safety model, connector health diagnosis, native context ingestion. The system is more sophisticated than it appears from the outside.

4. UX

Press, speak, done. Behind it: SSE streaming, sentence-level TTS, action review flows, deep-link orchestration, proactive briefing feeds, native local actions, BLE pendant bridging, bone conduction output. The UX is the product.

5. Trust

Press-to-talk. No camera. Visible memory. Review-gated high-risk actions. Per-user auth. These are not features — they are the trust architecture that makes the product safe to wear every day.

6. High-Agency AI

Most AI assistants answer questions. Milgrain executes tasks. Gmail send, Calendar create, Telegram to a real contact, Uber with your current location, trains with actual departure times, food orders with review. A 20-second task completed without the phone leaving the pocket.

12. Security & Privacy

12.1 Implemented

- Per-user auth: registration, login, password hashing (scrypt + random salt), 7-day signed session tokens.
- Auth required on most API routes. User ID matching prevents cross-user data access.
- CORS restricted via APP_URL. CSP and HSTS headers set.
- High-risk action review queue — user confirms before execution.
- Audio recording only on press-to-talk. No ambient listening. No camera.

12.2 Needs attention before real users

- OAuth token encryption at rest — currently stored as JSONB in Supabase.
- Supabase RLS policies — row-level security not confirmed in migration files.
- Debug endpoint must be protected or removed in production.
- Data retention / deletion policy — conversation history grows quickly.
- Privacy policy — not yet written.
- Rate limits — currently only on /process-audio; all other routes need limits.
- Waterproofing strategy for health sensors.

13. Roadmap

13.1 Immediate — June/July 2026

- Verify backend deploy end-to-end: /chat, /process-audio, /companion-live.
- Benchmark Gemini Live path vs upload/SSE on real device.
- Implement CoreBluetooth BLE pendant bridge in iOS app.
- ESP32-S3 prototype: mic test, bone conduction test at nape, BLE bridge on bench.
- Resin/3D printed countermodule + pendant dummy — on-neck scale and comfort test.
- Bone conduction acoustic test at nape against skin.

13.2 August 2026

- TestFlight: 50–100 users. Daily-use loop, proactive briefings, native actions, connectors.
- Demo video: phone stays in pocket, real actions execute, fashion object centre-frame.
- Landing page and Kickstarter page live.
- Product renders: pendant + countermodule, shell swap, jewellery-box charger.

13.3 September–December 2026

- Kickstarter launch — convert TestFlight users.
- nRF52840 PCBs designed in KiCad (pendant + countermodule).
- JLCPCB PCBA first run.
- Titanium shell CNC sample — one piece before batch quote.
- Conductive chain supplier and specification locked.
- PCBA and CNC quotes obtained — real unit economics.

13.4 2027

- Kickstarter fulfilment.
- Subscription conversion from hardware users.
- Shell drops as recurring revenue and cultural events.
- On-device intent classification (nRF52840 TFLite Micro).
- Health features: readiness score, cycle tracking, sleep coaching.
- Angel round only from demonstrated traction.

14. Key Open Questions

Bone conduction at nape	Needs physical testing — coupling through clothing vs bare skin, volume levels.
Conductive chain spec	Contact resistance, current capacity, flexibility, aesthetic. No supplier locked yet.
Countermodule form	Dimensions, enclosure material, wear comfort at nape across body types.
Health sensor contact	PPG needs skin contact — does the countermodule rest firmly enough throughout the day?
PCB Gerbers	Two boards needed. KiCad design has not started.
PCBA / CNC quotes	Unit economics are estimates until real quotes from JLCPCB and CNC supplier.
Gemini Live path	Benchmark on device before committing — upload/SSE is proven, Live is unverified at scale.
Token encryption	OAuth tokens in Supabase JSONB need encryption at rest before production users.
Waterproofing	No IP rating strategy. Sweat and rain will reach both modules at the neck.
Brand palette	May docs reference teal accent; current app leans warm stone/taupe. Pick one system.

15. One-Page Summary

What

Milgrain is a two-module wearable: a 20–24 mm titanium pendant at the front (mic, MCU, BLE) connected via a conductive chain to a countermodule at the nape (battery, bone conduction, health sensors). Press to talk. An AI listens, remembers you, and executes real tasks — Gmail, Calendar, Telegram, Uber, trains, food delivery, Apple Music, Maps, HomeKit. The phone stays in the pocket.

Why it's different

Fashion object, not gadget. 20–24 mm pendant is jewellery-scale. Swappable shells, jewellery-box charger, titanium and Gorilla Glass. No camera. Press-to-talk trust model. Real connector execution with review gates. Health sensing from the nape: HR, HRV, SpO2, temperature, sleep — Oura-equivalent from a better anatomical site. Memory that builds over time. Proactive briefings before you ask.

Business

Hardware at £349 (Kickstarter £299). Subscription at £14.99/month or £129–£149/year. Shell drops at £79–£89. At 1,000 units: ~£161 hardware margin per unit + ~£7–9/month subscription contribution. Hardware margin and shells are the core profit drivers at early scale.

Where it is now

Authenticated Cloud Run backend, PWA, SwiftUI iOS app, Gemini voice/chat/image, memory, proactive briefings, 15+ connectors and native actions, action safety model. Prototype firmware for Seeed XIAO nRF52840 streams audio over BLE. Hardware CAD, PCB Gerbers, conductive chain spec, CNC quotes, bone conduction acoustic test, and BLE iOS bridge are the major open items.

Next milestone

Physical prototype on-neck with bone conduction tested at the nape. TestFlight for 50–100 users by August. Kickstarter page live. Demo video: phone stays in pocket, real actions complete, fashion object and nape module centre-frame.

MILGRAIN

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